

Proposed Modification of the Paper Based on the New Real-Data Experiments

Paper: Business Location Recommendation with Geospatial Data and Generative AI

Purpose: integrate the student's real-data observations into the manuscript so that the paper answers reviewer criticism more directly, without overstating what the LLM contributes.

1. Main recommended change

Reframe the paper from “LLMs improve recommendation quality” to a more defensible claim: classical/CF and ML baselines are strong for pure top-N location recommendation, while LLMs add value mainly by extracting and enforcing natural-language constraints and by generating explanations. This directly answers criticisms about proxy labels, lack of baselines, and whether the LLM is really necessary.

The current paper already contains the core dataset, proxy-ground-truth construction, fine-tuned LLM versus RAG comparison, and a future-work paragraph on implicit-feedback CF baselines. The new experiments should move part of that discussion from “future work” into the main experimental evidence.

2. Where to integrate each observation

Observation from notebook/data	Manuscript modification	Why it helps reviewers
Section 4 - Baselines: RF/content-based classifiers with category encoding vs description embeddings; embeddings do not improve recommendation quality.	Add a new subsection in Section 5: “Classical, content-based, and embedding baselines”. Include a table comparing CF, RF-category, RF-description-embedding, RAG, and fine-tuned LLM.	Shows that the authors tested whether LLM-style semantic representations improve pure recommendation. The negative result is valuable and supports a narrower, more credible contribution.
Section 5 - Constraint satisfaction: pure CF recommends central areas regardless of user constraints; LLM/RAG captures nearly all constraints.	Add CSR@3 / CSR@10 as a new metric and a constraint benchmark with queries such as near port, quiet area, near university, away from center, near transport.	Directly demonstrates the unique value of LLMs beyond top-N accuracy: understanding and operationalizing user preferences.
Section 7 - Proxy sensitivity: different proxy weight sets produce stable results.	Extend Section 3.3 or Section 5 with “Proxy-weight sensitivity analysis”. Report Jaccard@10, Spearman rank correlation, and top-3 stability across weight schemes.	Answers criticism that the proxy labels are arbitrary or fragile.
Section 8 - Spatial robustness: merge neighborhoods into larger zones and compare results.	Add “Spatial-unit robustness” experiment: original 48 neighborhoods vs merged larger zones.	Answers the modifiable areal unit problem (MAUP): conclusions should not depend strongly on one neighborhood partition.
Section 9 - Feature analysis: business presence is mostly associated with centrality.	Add an “Ablation / feature diagnostic” paragraph. State that centrality is a dominant driver of observed business presence and explain why this creates a bias in CF/top-pop baselines.	Explains why pure CF tends to central areas and motivates constraint-aware re-ranking.

3. Proposed revised paper structure

Section 1 - Introduction: add one contribution: “We provide a real-data validation layer including CF/content-based baselines, constraint satisfaction evaluation, proxy sensitivity, and spatial-unit robustness.”

Section 3.3 - Proxy Ground Truth: keep the current ensemble construction, but add sensitivity analysis showing that rankings are stable under alternative weight sets. Make clear that the proxy is used for emulation, not as a claim of business success.

Section 4.2 - RAG Pipeline: explain that the full query is not only used for semantic re-ranking but also for constraint extraction. Add a small rule table mapping textual constraints to geospatial predicates.

Section 5 - Experimentation: split into five subsections: 5.1 top-N and content baselines; 5.2 fine-tuned vs RAG; 5.3 constraint satisfaction; 5.4 proxy sensitivity; 5.5 spatial robustness and centrality bias.

Section 6 - Conclusions: state the contribution more carefully: LLMs do not necessarily improve pure recommendation over strong CF/content baselines; their main contribution is natural-language constraint handling, explainability, and interactive personalization.

4. Tables/figures to add

New table/figure	Recommended content
Table A: Pure recommendation baselines	Rows: TopPop, ItemCF, PureSVD, NeuMF, RF-category, RF-description-embedding, Fine-tuned LLM, RAG. Columns: HR@K/NDCG@K or Match@3/Precision@3/Recall@3, plus long-tail subset.
Table B: Constraint Satisfaction Rate	Rows: Pure CF, RAG without constraint parser, RAG+LLM constraint extraction, Fine-tuned LLM. Columns: CSR@3, SoftCSR@3, examples of failed constraints.
Figure A: Proxy sensitivity	Heatmap or line plot of Jaccard@10 / Spearman correlation across weight sets.
Figure B: Spatial robustness	Bar chart comparing metrics for 48 neighborhoods vs merged zones.
Figure C: Centrality diagnostic	Feature importance or partial dependence showing centrality/distance-to-center as a dominant predictor of business presence.

5. Draft text to insert in the paper

Add to end of Introduction contributions:

"We complement the LLM evaluation with a real-data validation layer that includes classical collaborative-filtering and content-based baselines, constraint-satisfaction evaluation, proxy-weight sensitivity analysis, and spatial-unit robustness tests. These experiments clarify the respective roles of algorithmic recommenders and LLM components: the former provide competitive pure top-N ranking, while the latter are most useful for extracting natural-language constraints and converting them into personalized, explainable recommendations."

Add after current Section 5 metrics:

"To evaluate whether the system respects user-stated spatial preferences, we introduce Constraint Satisfaction Rate (CSR@k). For a query with hard constraints C and a ranked list of k recommended locations, CSR@k is the fraction of recommended locations satisfying all constraints. SoftCSR@k averages graded satisfaction scores for soft constraints. Textual constraints such as near the port, close to the university, away from the city center, quiet area, or close to public transport are mapped to explicit geospatial predicates using the available distance and accessibility features."

Add to Results/Discussion:

"The content-based Random Forest baselines indicate that replacing a simple business-category encoding with description embeddings does not improve pure recommendation quality on the current data. This negative result is informative: semantic business descriptions alone do not appear to solve the core spatial ranking problem. Instead, the main empirical advantage of LLM-based processing emerges in constraint extraction and constraint-aware re-ranking, where pure collaborative filtering tends to reproduce historically central business concentrations even when the user expresses a different spatial preference."

Add to Limitations:

"Observed business presence is strongly associated with centrality. Consequently, models trained only on historical co-location patterns may over-recommend central areas. This is not only a limitation but also a diagnostic result: it explains the failure mode of pure CF/top-pop baselines and motivates the hybrid architecture in which a recommender backbone proposes candidates while an LLM-driven layer filters or re-ranks them according to explicit user constraints."

6. Recommended response to the student

Ask for the exact numeric results and artifacts, not just the qualitative summary. Specifically request: (i) tables of metrics, (ii) definition of merged zones, (iii) constraint-test query set, (iv) examples where CF fails and LLM succeeds, and (v) code/output files for reproducibility.

7. Bottom-line recommendation

Yes, these observations should be integrated. The most important addition is the constraint-satisfaction experiment, because it gives the paper a clearer and stronger answer to criticism: LLMs are not claimed to magically outperform all recommenders; rather, they make the system usable in realistic conversational settings where entrepreneurs express constraints in natural language.